

A12749W1

SECOND PUBLIC EXAMINATION

Honour School of Philosophy, Politics, and Economics
Honour School of Economics and Management
Honour School of History and Economics

MICROECONOMICS

OUTLINE ANSWERS TO PROBLEMS

TRINITY TERM 2020

Wednesday 27 May, 09:30–12:30

Please start the answer to each question on a separate page.

There are 10 questions in this paper.

*Answer **all** questions from Part A and **two** questions from Part B.*

Part A attracts 40% of the marks in total; each question in Part B attracts 30% of the marks.

*In Part A, the weight assigned to each question (as a percentage of the total for Part A) is indicated in square brackets;
in Part B, the weight assigned to each part of the multipart question is indicated in square brackets.*

Candidates may use their own calculators.

Do not turn over until told that you may do so.

Part A

In Part A, the weight assigned to each question (as a percentage of the total for Part A) is indicated in square brackets

1. Consider an exchange economy with two consumers and two types of good. Consumer A has preferences represented by the utility function $u^A(x^A, y^A) = 2 \ln x^A + \ln y^A$, and an endowment of $(4, 4)$ (that is, 4 units of each good); B has preferences represented by the utility function $u^B(x^B, y^B) = \ln x^B + 2 \ln y^B$, with an endowment of $(11, 8)$ (that is, 11 units of good x and 8 units of good y).
 - (a) Explain why the endowment is not Pareto efficient.
Illustrate this situation using an Edgeworth box.
 - (b) Suppose that good y cannot be transferred between the consumers.
Find the transfer (of good x) that would result in a Pareto efficient allocation.
Mark this allocation on your diagram.
 - (c) Can the initial allocation and the allocation in part (b) be ranked using the Pareto criterion? Explain why or why not.
 - (d) Now suppose that good y can also be transferred between the consumers.
Show that the allocation that would be chosen by a social planner whose objective was to maximise the sum of the utilities of the two consumers is the same as the efficient allocation from part (b).
Is this different from the allocation that a social planner would choose if their objective was to maximise the minimum of the utilities of the two consumers?
How do you know?

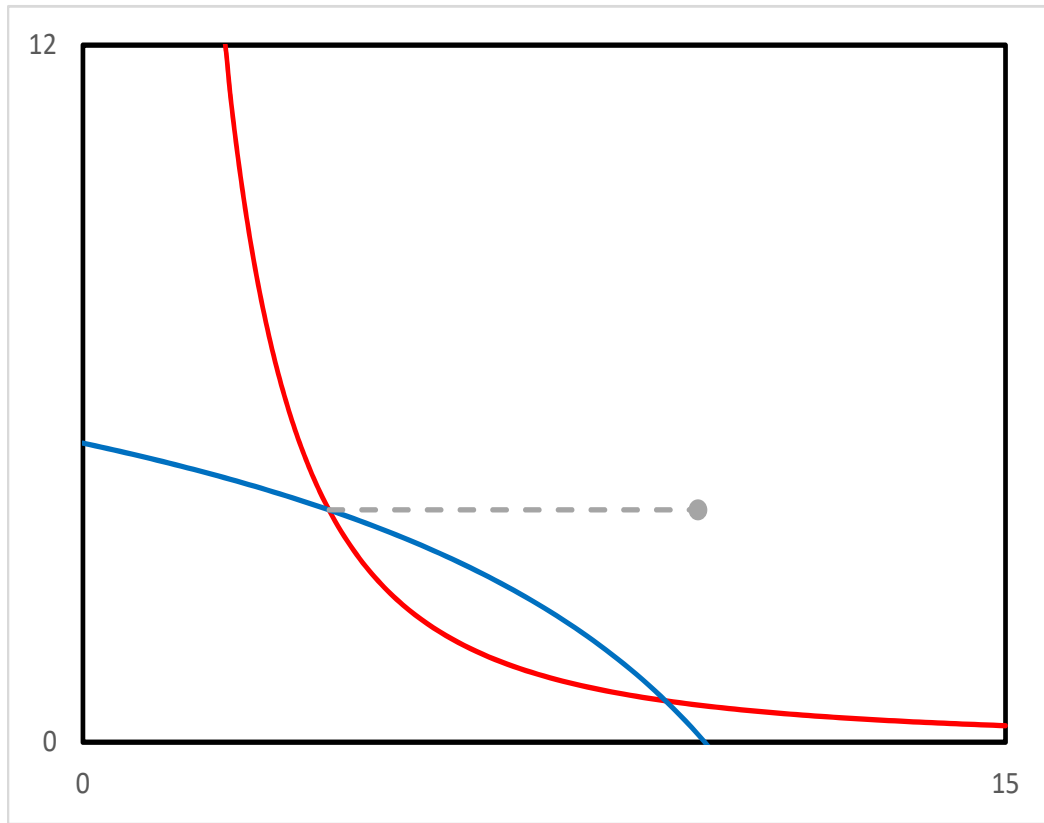
[25%]

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1. For MRS & MRT, negative values are also acceptable

- (a) A condition for efficiency is equal marginal rates of substitution.
 $MRS^A = 2y^A/x^A$, $MRS^B = y^B/2x^B$;
 evaluate at $(4, 4)$ and $(11, 8)$ to give $MRS^A = 2$, $MRS^B = 8/22$ which are not equal.
 Therefore inefficient.
- (b) For efficiency we require equal MRS's, and the endowment to be exhausted.
 Therefore need $2y^A/x^A = y^B/2x^B$, so $8/x^A = 8/2(15 - x^A) \Rightarrow x^A = 10, x^B = 5$.
 Transfer 6 units of x from B to A : A will have $(10, 4)$, B will have $(5, 8)$.
- (c) A has become better off, B has become worse off.
 Together, these imply that the two allocations are not Pareto comparable.
- (d) SP would choose (x^A, y^A) to maximise $2 \ln(x^A) + \ln(y^A) + \ln(15 - x^A) + 2 \ln(12 - y^A)$.
 $FOC_x \Rightarrow 2/x^A = 1/(15 - x^A)$, so $x^A = 10$;
 $FOC_y \Rightarrow 1/y^A = 2/(12 - y^A)$, so $y^A = 4$;
 these are the same as in part (b).

$u^A = 2 \ln 10 + \ln 4 = \ln 400$, $u^B = \ln 5 + 2 \ln 8 = \ln 320$;
 if SP were maximising the minimum, then $u^A = u^B$, which is not the case here.

Edgeworth box



2. Two colleges, Larches and Osier, are currently equal in the academic league table, but each is considering extra coaching for their students. They simultaneously decide whether to *Maintain* current levels, M , or *Increase* the level of coaching, I , payoffs being given by:

		LARCHES	
		<i>Maintain</i>	<i>Increase</i>
OSIER	<i>Maintain</i>	0 2	−1 −4
	<i>Increase</i>	−1 −4	2 −4

- (a) Define the Nash equilibrium of a game.

Find all the Nash equilibria of the one-shot game above (in both pure strategies and mixed strategies, if any).

Calculate the expected payoffs of each college in each of the equilibria.

- (b) Explain what is meant by a *subgame perfect equilibrium* of a repeated game.

- (c) Suppose that the two colleges repeat the above stage game infinitely often.

Show that $(\textit{Maintain}, \textit{Maintain})$ in every period can be sustained as the outcome of a subgame perfect equilibrium for a discount factor sufficiently close to one.

[*Hint*: consider a grim trigger strategy with the threat of reverting to a suitable Nash equilibrium of the stage game from part (a).]

[25%]

2. (a) NASH EQUILIBRIUM (NE): a strategy profile $(s_1^*, s_2^*, \dots, s_K^*)$ such that s_k^* is a best response to s_{-k}^* for each player $k = 1, 2, \dots, K$.

NE: (M, I) , (I, M) , and $(\frac{3}{5}M + \frac{2}{5}I, \frac{3}{5}M + \frac{2}{5}I)$.

Expected payoff in the mixed NE is $-\frac{2}{5}$ each.

- (b) SUBGAME PERFECT EQUILIBRIUM (SPE): a NE which induces a NE in each subgame.

- (c) Consider the grim trigger strategy: play M unless either player has played I ; then play as in the mixed NE from part (a) forever, starting in the period immediately after the deviation.

‘Co-operation’ yields 0.

‘Deviation then punishment’ yields $2 + \delta/(1 - \delta) \times (-\frac{2}{5})$.

SPE if $0 \geq 2 + \delta/(1 - \delta) \times (-\frac{2}{5}) \iff \delta \geq 5/6$.

{>>> Note: Alternative punishment is to revert to the NE disliked by the deviator. Then the condition for SPE is $0 \geq 2 + \delta/(1 - \delta) \times (-1) \iff \delta \geq 2/3$.}

3. (a) Define the certainty equivalent of a lottery, and explain how it can be used to determine whether or not someone should participate in a given lottery.

Janet is an expected utility maximiser who prefers more to less, and whose broad attitude to risk (risk averse, risk neutral, or risk loving) is independent of her wealth.

There are two projects. If Janet invests in project X , her final wealth will be either 6 or 30, each equally likely; if she invests in project Y , her final wealth will be either 12 or 20, again each equally likely. She is prepared to invest in one or the other, but is not allowed to invest in both.

- (b) She tells her friend Sam that she prefers Y to X , from which Sam infers that Janet is risk averse, reasoning as follows:

Consider project Z which would leave Janet with either 14 or 22, each equally likely. She would certainly prefer Z to Y , and if she were risk neutral or risk loving then she would be content with X instead of Z , and consequently prefer X to Y – but she doesn't, so she must be risk averse.

Evaluate Sam's argument.

- (c) Janet now hears that the cost of investing in project Y is going to increase, effectively reducing her final wealth. If her utility from final wealth w is $\ln w$, what is the largest cost increase Janet would find acceptable, above which she would prefer to invest in project X ?

[25%]

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3. (a) The certainty equivalent is y^c where $u(y^c) = \bar{u}$, where $\bar{u}$ is the expected utility from taking part in the lottery.

An EU maximiser would take part in a lottery if and only the certainty equivalent is at least as large as final wealth from NOT taking part.

- (b) {>>> An argument just using 'variance' instead of 'spread' is invalid.}

For Janet: $\frac{1}{2}u(12) + \frac{1}{2}u(20) \geq \frac{1}{2}u(6) + \frac{1}{2}u(30)$.

For an EU maximiser: $\frac{1}{2}u(14) + \frac{1}{2}u(22) > \frac{1}{2}u(12) + \frac{1}{2}u(20)$ (+2 in each state);

if risk neutral: $\frac{1}{2}u(6) + \frac{1}{2}u(30) = \frac{1}{2}u(14) + \frac{1}{2}u(22)$ (same mean);

if risk loving: $\frac{1}{2}u(6) + \frac{1}{2}u(30) > \frac{1}{2}u(14) + \frac{1}{2}u(22)$ (mean preserving spread).

Given that her broad attitude to risk is independent of her wealth – and this is CRUCIAL to the above argument going through – Janet cannot be either risk neutral or risk loving, so Sam's reasoning is correct.

- (c) To make J indifferent: $\frac{1}{2}u(12 - p) + \frac{1}{2}u(20 - p) = \frac{1}{2}u(6) + \frac{1}{2}u(30)$

With $u(w) = \ln w$:

$$\begin{aligned}
 (12 - p)(20 - p) &= 6 \times 30 \\
 \Rightarrow p^2 - 32p + 240 &= 180 \\
 \Rightarrow p^2 - 32p + 60 &= 0 \\
 p &= \frac{32 \pm \sqrt{32^2 - 4 \times 60}}{2} = 16 \pm \sqrt{256 - 60} \\
 &= 16 \pm 14 \\
 p &= 2
 \end{aligned}$$

4. There are two types of worker: H -types with high productivity θ_H , and L -types with low productivity θ_L ; each worker knows their own type, but they have no outside option. Many risk-neutral firms compete for the services of the workers, but they cannot observe a worker's type.

- (a) Why would some of the workers like to be able to signal their type to the firms?
What condition(s) would make a signal credible?

Suppose that $\theta_H = 500$, $\theta_L = 400$, and that $3/5$ of the workers are H -types, so $2/5$ of the workers are L -types; the firms are aware of these productivity levels and proportions.

- (b) When no signal is available, what is the market equilibrium outcome?
(c) Suppose the workers can acquire an education, observable by the firms – this would cost 80 for an L -type, but only 60 for an H -type.

Consider an outcome in which H -types signal and L -types don't, and the firms pay appropriate wages.

Is this outcome a market equilibrium? Explain why or why not.

If not, find an alternative outcome, and show that it is an equilibrium.

Is the equilibrium efficient?

[25%]

4. (a) H -types would like to signal and get a wage $w = \theta_H$ rather than average productivity, if the cost of the signal is less than the wage difference.

To be credible, signal must be too expensive for L -types to copy.

- (b) Expected productivity: $\frac{3}{5}\theta_H + \frac{2}{5}\theta_L = \frac{3}{5} \times 500 + \frac{2}{5} \times 400 = 300 + 160 = 460$.

All workers are employed, at a wage of 460.

- (c) Separating is not an equilibrium:

if the firm pays $w_H = 500$ to a worker that signals, and $w_L = 400$ to one that doesn't, then L -types would mimic, since $500 - 80 = 420 > 400$.

{>>> **NOT:**

H -types have a profitable deviation to pooling and receiving $460 > 500 - 60$!!!

– the pooling wage is not available in the conjectured separating equilibrium.}

The workers not signalling and then receiving a wage of 460 is a pooling equilibrium.

(H -types wouldn't deviate – at best they would net is $500 - 60 = 440 < 460$.)

In this case, the equilibrium is efficient and is the same as when signalling is unavailable.

{>>> There are also two semi-separating equilibria (neither expected nor required):

- L 's don't signal, H 's don't signal w.p. α ;

H 's are willing to mix when expected productivity = $\theta_H - c_H$, so $\alpha = 4/9$;

- H 's signal, L 's signal w.p. β ;

L 's are willing to mix when expected productivity = $\theta_L + c_L$, so $\beta = 3/8$.}

Part B

10. Consider a duopoly in which two firms produce goods that are perfect substitutes, and they compete for one period only, simultaneously choosing their production level. The cost to firm j of producing a quantity $q_j \geq 0$ is $15q_j$, for $j = 1, 2$, and the inverse demand function in the market is

$$p(q) = 75 - q \quad \text{where } q = q_1 + q_2.$$

- (a) Find the pure-strategy Nash equilibrium of this game.

What is the resulting price, and how much profit does each firm make? [20%]

Suppose that before the two firms play the above quantity-setting game, firm 1 can make a cost-reducing investment that has a lump-sum price of $K > 0$ and results in its cost function becoming $9q_1$. Firm 2 knows whether or not firm 1 made the investment before the firms choose how much to produce.

- (b) If firm 1 decides to invest, what will be the Nash equilibrium of the quantity-setting game?

What is the resulting price, and how much profit does each firm make (ignoring the prior investment cost of firm 1)?

What are the reasons why firm 1 earns more than in part (a) while firm 2 earns less? [30%]

- (c) Consider the two-stage game in which firm 1 chooses whether to invest or not in the first stage, and, having noted the investment decision of firm 1, the firms engage in quantity competition in the second stage.

In a subgame-perfect Nash equilibrium of the above two-stage game, what is the largest value of K that would result in firm 1 making the investment?

When K is low enough for firm 1 to make the investment, describe fully the strategies of the two firms in the subgame-perfect Nash equilibrium. [25%]

- (d) Now suppose $K = 150$, but that before the game starts, firm 2 announces that at the second stage it will ignore firm 1's investment decision and produce its quantity from the equilibrium in part (a).

If firm 1 believed the announcement, would it invest?

Should firm 1 invest – that is, is firm 2's announcement credible? Explain. [25%]

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10. (a) Best response is $q_i = (75 - 15 - q_j)/2$, for $i, j = 1, 2$.

Symmetric, so $q^* = (75 - 15)/3 = 20$.

Price: $p^* = 75 - 2 \times 20 = 35$; profit per firm: $\pi = 20^2 = 400$.

- (b) If firm 1 invests then best response is $q_2 = (60 - q_1)/2$ as before,
but now $q_1 = (75 - 9 - q_2)/2$.

The equilibrium is $q_1^K = 24$, $q_2^K = 18$.

Price: $p^K = 33$; profits: $\pi_1^K = 24^2 = 576$, $\pi_2^K = 18^2 = 324$.

Firm 1's production cost has fallen for any given level of output, and this makes it more aggressive, pushing out its best response curve for any q_2 .

In this Cournot setting, choices are STRATEGIC SUBSTITUTES:

firm 1's equilibrium production increases, firm 2's decreases.

Aggregate quantity increases, pushing down the price (harming firm 2),
but this is more than compensated for firm 1 by its increase in quantity.

- (c) $\bar{K} = 576 - 400 = 176$,

namely the difference between π_1^K and π^* from part (b) and part (a).

SPE:

Firm 1 invests at stage 1, then at stage 2 plays according to part (b), i.e. $q_1^K = 24$.

Firm 2 plays according to part (b) if investment occurs, i.e. $q_2^K = 18$,

but according to part (a) if investment does not occur, i.e. $q_2^K = 20$.

{>>> Note: Some answers might specify what firm 1 would do at stage 2 had it not, counterfactually, invested at stage 1.}

- (d) If firm 1 believed firm 2 then it expects $q_2 = 20$ and hence plays $q_1' = (66 - 20)/2 = 23$ (instead of $q_1^K = 24$).

Price is $p' = 32$ and then $\pi_1' = 23^2 = 529$ so NET profit would be $529 - 150 < 400$ – not worth investing.

The threat is not credible – if firm 1 were to invest, then firm 2 would do better by not carrying through its ‘threat’ – so firm 1 should invest.
